

1. in 2D rotate the axes clockwise by 45 degrees. Where will the points appear in new system (2,3), (3,2), (-1,-1). verify by using the matrix
2. if we do the following transformations one by one, we should get original system back, verify that its true by multiplying both transformation matrices, you should get I matrix
 - (I) rotate x axis clock wise by angle A
 - (II) rotate x axis anticlock wise by angle A
3. in scaling in 2D in both directions we can do them simultaneously or we can do it one by one first X direction, then Y direction. You can verify using multiplying Tms. Verify if it works for shearing, is shearing in both directions simultaneously same as shearing in X, then shearing in Y direction?
4. find out the TM needed to do projection on all points in 2d onto a line $y=mx$. You can do it in 2 ways, do both
 - (I) naive way I mentioned in class, find (h,k) for every (x,y) in terms of x,y,m
 - (ii) first shift x axis along the direction of $y = mx$, then scaling along Y direction with factor = 0. this will collapse all points onto the new X axis, and rotate the X axis back to original position
 - (iii) verify both are giving same TM or not
5. write a TM to do projection onto a plane $ax+by+cz=0$ in 3D. Just find the foot of perpendicular of every point (x,y,z).
6. write a TM to do the following transformation : rotate every point in XY plane by A degrees in anticlockwise around the point (a,b). verify the TM you obtained is true with following cases
 - (I) A is 90 degrees, (a,b) is (4,4). verify for the point (8,4), and (0,0). first from the graph try to identify where they'll shift. Then verify by substituting values in the TM